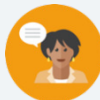


14.10 Solving One-Step Inequalities Using Multiplication and Division

Lesson Introduction

 Benchmark	MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.		 Learning Targets	- I can determine the maximum or minimum value that makes an inequality true. - I can explore the relationship between coefficients of the terms in an inequality and solution set. - I can solve one-step inequalities in one variable using multiplication or division. - I can relate properties of inequality to solving one-step inequalities algebraically.	
 Benchmark Coherence	Building On			Working Toward	
	MA.6.AR.1.2 Translate a real-world written description into an algebraic inequality in the form of $x > a$, $x < a$, $x \geq a$, or $x \leq a$. Represent the inequality on a number line. MA.6.AR.2.1 Given an equation or inequality and a specified set of integer values, determine which values make the equation or inequality true or false.			MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.	
 Essential Questions	- What is the minimum or maximum value that makes the inequality true? - How do you relate the solution set of an inequality to the context it represents? - How do you determine if solutions are reasonable for the context? - What is the effect multiplying and dividing by a negative number has on the inequality? - How do I use properties of inequalities to solve one-step multiplication and division inequalities?				
 Lesson Narrative	In this lesson, students read, graph, and interpret inequalities in the form $px < q$ and $\frac{x}{p} < q$. Students explore the effect multiplying and dividing an inequality has on the inequality. Students explore the meaning of a minimum and maximum value in terms of inequalities and connect these to a real-world context. Students use properties of inequality to solve one-step multiplication and division inequalities. Students also graph the solution sets and test values to prove their solutions are true.				
 Component Summary	14.10.1 Warm-Up (~5 min.)	Write and interpret inequalities based on hanger diagrams (SE p. 391)	14.10.4 Collaboration (10 min.)	Explore the effect of a negative coefficient on an inequality (SE p. 394)	
	14.10.2 Exploration (15 min.)	Explore maximum and minimum values of inequalities (SE p. 392)	14.10.5 Bring It Together (5 min.)	Summarize the effect of a negative coefficient on an inequality (SE p. 395)	
	14.10.3 Bring It Together (5 min.)	Relate the solution set to the context it represents (SE p. 394)	14.10.6 Guided Instruction (10 min.)	Solve one-step inequalities with multiplication and division (SE p. 396)	
	14.10.7 Wrap-Up (~5 min.)	Compare the process of solving inequalities with positive and negative coefficients (SE p. 397)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Mathematical models - Multiplication property of inequality - Division property of inequality <u>Additional Terms:</u> - Solution set - Minimum - Maximum		(Optional) Calculators (Optional) Chart paper (Optional) Markers		- Consider creating an anchor chart with instructions, examples, and visuals for solving one-step inequalities with multiplication and division. - Consider editing the real-world problems based on the interests of your students.

 Teacher Tips	Common Misconceptions - Students may struggle to read and interpret inequalities with the variable on the right side. - Students may not realize that $-x$ has a coefficient of -1. - Students may struggle to interpret the solution set in the context of the problem.	Things to Consider - Consider creating and displaying an anchor chart with vocabulary definitions, instructions, examples, and visuals. - Consider asking students to practice reading inequalities from right to left and left to right. - Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
	Literacy and Language Routines Think-Pair-Share Use 14.10.4 Collaboration question 1 as an opportunity for students to initially try the problems independently and then discuss their strategies and answers with their partner. These discussions will help students gain confidence and understanding.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.6.1: Reasonableness In 14.10.2 Exploration, when determining the solution set for an inequality, ask students to consider if the solution makes sense in the context they represent. Students discover that within a real-world context, often a negative value does not make sense.
	 Differentiate Instruction	<i>For the 14.10.7 Wrap-Up, students are asked to write an explanation of the similarities and differences between solving inequalities with positive and negative coefficients. If students struggle expressing themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.</i>
Lesson Notes:		

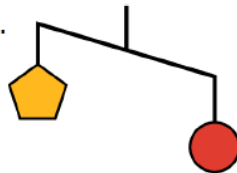
14.10.1 Warm-Up: Visual Models

Component Overview: Students write and interpret inequalities based on hanger diagrams. Consider first modeling how to write the description and inequality. Then ask students to work independently or with a partner. Explain to students that p , c , s , and t represent the weights of the pentagon, circle, square, and triangle, respectively.



14.10.1 Warm-Up: Visual Models

An unbalanced hanger diagram is shown. It shows that the weight of one circle is greater than the weight of one pentagon.



1. Complete the table.

Hanger Diagram	Description	Inequality
	The weight of one circle, c , is greater than the weight of one pentagon, p .	$p < c$
	The weight of one pentagon, p , is greater than the weight of one square, s .	$s < p$
	The weight of one square, s , is <i>less</i> than the weight of a circle and a pentagon.	$c + p > s$
	The weight of one square, s plus the weight of one circle, c is greater than the weight of two triangles, t .	$s + c > 2t$



As students complete the Warm-Up, circulate the room, listening for student misconceptions. Consider creating a chart to make notes that may inform remediation.



- Is there another way you could write the inequality $p < c$?
- Can you read the inequality from right to left?

14.10.2 Exploration: Determining the Minimum or Maximum Value That Makes an Inequality True

Component Overview: Students explore how to determine the maximum and minimum values of inequalities. Students should use substitution to determine if a value will satisfy the inequality. Students also determine which solutions make sense in the context of the problem.



14.10.2 Exploration: Determining the Minimum or Maximum Value That Makes an Inequality True

Work with your partner to complete the following.

1. Bak Middle School of the Arts in West Palm Beach, FL is hosting a rehearsal for its upcoming musical. The stage manager has \$83 from a recent fundraiser that can be used to order lunch for the cast and crew. The school's cafeteria can make boxed lunches for a cost of \$6 each. The inequality $6n \leq 83$ represents the situation, where n represents the number of boxed lunches that can be ordered from the cafeteria.
 - a. Select the values that represent a valid number of boxed lunches the stage manager can order from the cafeteria.

☐ 14	☐ 9.5
☐ $13\frac{5}{6}$	☐ 2
☐ 12	☐ -4
 - b. Explain why some values in the previous question are not valid in this context.
14 is too large of a number and -4 does not make sense in this context.
 - c. The mathematical solution to the inequality $6n \leq 83$ is $n \leq 13\frac{5}{6}$. What is the maximum number of boxed lunches the stage manager can order from the cafeteria?
13 is the maximum number because the cafeteria may not sell $\frac{5}{6}$ a lunch.
 - d. What is the minimum number of boxed lunches the stage manager can order from the cafeteria?
The least the stage manager can order is 0 boxed lunches.
 - e. The graph of the solution set for $6n \leq 83$ is shown.



Remind students that they can select multiple values for question 1a.



Encourage students to substitute each value into the inequality to test if it is true. For question 1b, consider highlighting the fact that -4 would not make sense in the context of the problem.

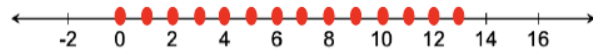


- What is the difference between maximum and minimum?
- How can you determine the maximum in this situation?
- How can you determine the minimum?
- For question 1c, should you round up or down? Why?

14.10.2 Exploration: Determining the Minimum or Maximum Value That Makes an Inequality True (continued)

Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line shown.

This is not a good representation because you cannot buy a negative number of lunch boxes.



2. Hiromi walks around his neighborhood each afternoon, tracking his pace each day. The fastest rate Hiromi can walk for an extended period of time is 3.7 miles per hour.

- a. What is the maximum distance Hiromi can walk in 0.75 hours?

2.775 miles

- b. The inequality $\frac{d}{3.7} \leq 0.75$ can be used to represent the distance Hiromi walks, d , in 0.75 hours given his fastest walking rate of 3.7 miles per hour.

Use your answer to Part A to determine the solution set for $\frac{d}{3.7} \leq 0.75$.

$d \leq 2.775$ miles

- c. The graph of the solution set for $\frac{d}{3.7} \leq 0.75$ is shown.



Discuss with your partner how the graph that mathematically represents the solution set is not the best representation of the real-world context. Then, represent the solutions in the real-world context on the number line.

This is not a good representation because you cannot run negative miles.



Consider providing students with the following fill-in-the-blank number sentence:

_____ $\times$ _____ = _____.



- If the maximum distance Hiromi can walk is 2.775 miles, how can we write that using an inequality?
- If 2.775 is the maximum, which way should the inequality symbol be facing?
- Should 2.775 be included or not included in the solution?



Think-Pair-Share

For question 2c, have students think independently first before sharing with a partner. Ask students to consider whether negative numbers are possible in this context.

14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Component Overview: Students relate solution sets to the contexts they represent. Ask students to look back at the questions from 14.10.2 Exploration and consider if any values should be excluded based on the context of the problem.



14.10.3 Bring It Together: Solutions and Mathematical Models in Context

Each of the inequalities in the Exploration has a mathematical solution set that can be represented on a number line. These graphs are examples of mathematical models.



Mathematical models are helpful tools used to understand a situation and make predictions. However, sometimes the mathematical model can include parts that do not make sense in a real-world context. When interpreting models to solve problems, it is important to analyze whether aspects of the model make sense in a given context.



Consider utilizing some time in this section to address misconceptions that students had during the previous 14.10.2 Exploration.

14.10.4 Collaboration: The Effect of a Negative Coefficient

Component Overview: Students explore and discuss the effects of negative coefficients on inequality symbols by comparing inequalities with positive and negative coefficients. Remind students that they can always see a visual representation of the inequality by graphing it on a number line.



14.10.4 Collaboration: The Effect of a Negative Coefficient

- Two simple inequalities are shown.

$$\textcircled{1 > 0} \text{ and } \cancel{1 > 0}$$

- Circle the inequality that is true and cross out the inequality that is false.
- Complete the statement.
If 1 is greater than 0, then the opposite of 1 is less than 0.
- Write the correct inequalities by filling in a comparison symbol on each line.
 $0 < 1$ and $0 > -1$
- Verify your answers with your partner on parts A-C and, if necessary, resolve any differences.

- Work with your partner to complete the following.
 - Complete the inequalities in the table based on your understanding of opposites.

Given Inequality	Equivalent Inequality
$0 < -q$	$0 > q$
$0 > -m$	$0 < m$
$-w \leq 0$	$w \geq 0$

In each equivalent inequality, the sign of the variable changed to its opposite and the value 0 remained the same.

- Describe which operation(s) could be applied to both sides of the given inequalities above to result in the signs of the variables changing and the value 0 remaining the same.
Multiplication or division by -1
- Describe what happens to the comparison symbol in the inequality when either of these operations are applied.

The sign flips or changes to the opposite direction.



Prompt students to practice reading inequalities from right to left and left to right. This practice will help students who have a hard time reading or interpreting inequalities with the variables on the right.



Think-Pair-Share

Use this routine for question 1d. Ask students to justify their answers to their partners. These discussions will build student confidence and understanding.



Why is it possible that both multiplying by -1 to both sides and dividing by -1 on both sides result in the same solution?

14.10.5 Bring It Together: The Effect of a Negative Coefficient

Component Overview: Students consolidate their understandings by summarizing the effect of a negative coefficient on an inequality.



14.10.5 Bring It Together: The Effect of a Negative Coefficient

- Complete the statements using words from the bank shown. Some words may not be used.

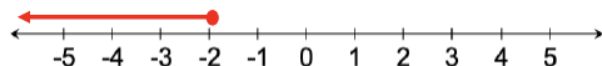
adding	subtracting	multiplying
dividing		opposite

When the variable term in an inequality is negative, the inequality can be rewritten by multiplying or dividing the inequality by -1 . In the resulting inequality, each term and the comparison symbol will change to its opposite.

- Use the inequality $-n \geq 2$ to complete the following.
 - Complete the inequality to create an equivalent inequality.

$$n \boxed{\leq} -2$$

- Graph the solution set for $-n \geq 2$.



ELL students may benefit from visual representations for each word in the word bank.



Which inequality, $-n \geq 2$ or $n \leq -2$, is easier to graph on a number line?

14.10.6 Guided Instruction: Solving One-Step Inequalities With Multiplication and Division

Component Overview: Students are guided through algorithms for solving one-step inequalities using the multiplication and division properties of inequalities.



14.10.6 Guided Instruction: Solving One-Step Inequalities with Multiplication and Division

As with solving one-step inequalities with addition and subtraction, properties of inequality can be used to solve inequalities with multiplication and division. As discovered in the Collaboration, though, when the value that is being multiplied or divided is negative, this changes both the values in the inequality *and* the comparison symbol to their opposites.

Multiplying both sides of an inequality by the same value can be used to solve an inequality by the **multiplication property of inequality**.



Multiplication Property of Inequality

If $a > b$ and $c > 0$, then $a \times c > b \times c$.

If $a > b$ and $c < 0$, then $a \times c < b \times c$.

- Find the solution to both inequalities using the multiplication property of inequality.

$\frac{y}{4} \leq 2.1$ $4 \times \frac{y}{4} \leq 2.1 \times 4$ $y \leq 8.4$	$\frac{y}{-4} \leq 2.1$ $-4 \times \frac{y}{-4} \leq 2.1 \times -4$ $y \geq -8.4$
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Dividing both sides of an inequality by the same value can be used to solve an inequality by the **division property of inequality**.



Division Property of Inequality

If $a > b$ and $c > 0$, then $a \div c > b \div c$.

If $a > b$ and $c < 0$, then $a \div c < b \div c$.

- Find the solution to both inequalities using the division property of inequality.

$-25 > 5c$ $\frac{-25}{-5} > \frac{5c}{-5}$ $-5 > c$	$-25 > -5c$ $\frac{-25}{-5} > \frac{-5c}{-5}$ $5 < c$
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Note that c cannot equal 0 for the multiplication property of inequality. That is why the property states c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (For the multiplication property, c cannot equal 0 because the two expressions would be equal, or both be 0, so the inequality statement would no longer be true.)



Consider initially presenting the properties using concrete examples (with numbers). After understanding the concrete examples, students will be ready for the abstract generalizations presented with variables.



Note that c cannot equal 0 for the division property of inequality. That is why the property states that c is either positive or negative.

Ask students to discuss with their partners why c cannot equal 0. (Since division by 0 is undefined, it is not possible to divide either expression by 0.)

If students forget to reverse the direction of the inequality symbol when multiplying by -5 , prompt them to check their work by substituting in a value for c . This approach will empower students to self-regulate as they identify and correct their own mistakes.

14.10.7 Wrap-Up: Cool Down

Component Overview: Students demonstrate their mastery of the lesson learning targets by explaining the similarities and differences between solving inequalities with positive and negative coefficients. Consider using data from this formative assessment to determine which students need remediation before the next lesson.



14.10.7. Wrap-Up: Cool Down

Two inequalities are shown.

$2.5x \geq -10$ $-2.5x \geq -10$

- 1. Complete the table to describe how the process of solving each inequality is similar and different.

How is the process of solving each inequality similar?	How is the process of solving each inequality different?
You must divide by the coefficient on both sides.	When dividing by the coefficient in the second equation the inequality sign must change to the opposite direction.



If students struggle to express themselves in writing, consider allowing them to circle the steps that are similar and underline the steps that are different and use an audio recorder or speech-to-text software.